RPG EXCELLENCE
MANAGEMENT SYSTEMS & BUSINESS ASSURANCE
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DECODING

ISO/IEC 17025:2017

A practical guide to laboratory competence, reliable results and accreditation readiness

For testing laboratories, calibration laboratories, sampling organisations, technical managers, quality teams and internal auditors.

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The examples, suggested records and posters are practical aids, not additional normative requirements. Accreditation-body policies can impose supplementary requirements; applicability varies by scheme, discipline, territory and scope. Technical figures are illustrative and must be validated for each method.

How to use this book

Read the chapter for the relevant clause; translate its intent into a laboratory-specific process; test the process with a real sample, instrument or job; and then check that the evidence supports the reported result. Use the posters for staff conversations and the final checklists for internal review.

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01 The logic of laboratory competence

ISO/IEC 17025 is not simply a document-control system with a laboratory attached. Its central concern is whether a laboratory can produce technically valid results, consistently, without undue influence, and can demonstrate why those results deserve confidence. A neat quality manual is of little value if an analyst uses an unverified method, an unsuitable reference standard or an uncontrolled calculation.

A useful way to understand the standard is as a chain of confidence: customer requirement → competent people → suitable method → controlled conditions and equipment → traceable measurements → valid technical records → checked results → clear, authorised report. Each link has to work.

The four assurance questions

Competence: Who is authorised, and on what evidence?
Technical validity: How do we know the method and measurement system are fit for purpose?
Integrity: What protects the result from bias, error and unauthorised change?
Confidence: What independent checks would detect a wrong result?

2.1 Scope and applicability

The standard addresses organisations conducting testing, calibration, and sampling linked to subsequent testing or calibration. A laboratory may be small, multi-site, embedded in a manufacturer or operate at customer premises.

How to make it work

Define activities, sites, methods, ranges and intended claims precisely. Distinguish laboratory conformity from the narrower activities formally listed on an accreditation schedule.

Evidence worth retaining

An activity/scope matrix linking methods, location, responsible roles and current accreditation status.

Auditor’s lens

Common weakness: Using “accredited laboratory” as a blanket claim for every service.
Question to ask: Can we show exactly which activity is covered by the claim made to a customer?

2.2 Normative references and terminology

Terms such as verification, validation, decision rule and metrological traceability have technical meanings. Misusing them leads to wrong decisions and misleading reports.

How to make it work

Maintain a short laboratory glossary and ensure staff can distinguish a method check from full validation, and a calibration from a functional check.

Evidence worth retaining

Training examples, method review records and correctly used terminology in reports.

Auditor’s lens

Common weakness: Calling a manufacturer service sticker proof of traceability without reviewing the certificate.
Question to ask: What evidence links our terminology to the actual technical work?

4.1 Impartiality

Confidence depends on technical conclusions being protected from sales targets, internal politics, customer pressure and conflicts of interest. Risk assessment is ongoing, not an annual signature exercise.

How to make it work

Map relationships that could influence a result: bonus structures, production reporting lines, client-funded projects, staff consultancy and repeat testing after a failure. Set escalation and independent review controls.

Evidence worth retaining

Conflict declarations, impartiality risk log, management decisions and examples of independent review.

Auditor’s lens

Common weakness: A generic declaration that ignores real commercial pressure.
Question to ask: Who could benefit if this result changed, and what stops them influencing it?

4.2 Confidentiality

Customer data, results, methods and information received from third parties need controlled disclosure. Confidentiality extends beyond employees to contractors and others acting for the laboratory.

How to make it work

Define legal and contractual obligations; restrict LIMS access by role; manage external communication, public case studies, cloud storage and authorised release.

Evidence worth retaining

Contracts, confidentiality commitments, access review, incident records and disclosure approvals.

Auditor’s lens

Common weakness: Assuming a supplier’s standard terms adequately cover laboratory information.
Question to ask: Can we trace who viewed, exported or released a sensitive result?

5 Structural requirements

A laboratory needs clear legal accountability, management responsibility, defined activities and authority to stop or correct technically compromised work.

How to make it work

Map the laboratory within its parent organisation. Define authority for technical approval, method changes, quality oversight, nonconforming work and release of results; cover mobile and off-site activities.

Evidence worth retaining

Organisation chart, role authorisations, documented activity scope and escalation examples.

Auditor’s lens

Common weakness: An organisation chart without decision rights or cover arrangements.
Question to ask: Who can stop release of a questionable report when the manager is absent?

6.1 Resource planning

Resources are a technical control, not merely a budget item. Insufficient staff, space, equipment or supporting services can directly compromise validity.

How to make it work

Connect workload forecasts to staffing, maintenance, calibration, consumables and environmental capacity. Trigger review when demand or method scope changes.

Evidence worth retaining

Capacity review, resource plans, maintenance backlog, contingency decisions.

Auditor’s lens

Common weakness: Treating missed maintenance as an administrative delay.
Question to ask: What happens to result validity when throughput doubles?

6.2 Personnel competence

Training attendance does not establish competence. A laboratory must know what a person can perform, interpret, check and authorise.

How to make it work

Define role-specific criteria; use witnessed practical demonstrations, blind samples, calculation checks and periodic monitoring. Authorise by method or activity, not broad job title.

Evidence worth retaining

Competence matrix, practical assessments, supervised work, signed authorisations and reauthorisation triggers.

> **Auditor's lens**
>
> Common weakness: Authorising a new analyst solely after an online course.
> Question to ask: Show me the most recent independent evidence this person can perform this method correctly.

6.3 Facilities and environmental conditions

Conditions matter when they can influence a measurement, sample integrity or safety of operation. Limits should be based on technical need.

How to make it work

Identify critical environmental parameters, monitor at relevant points and times, set excursion responses, segregate incompatible activities and validate environmental controls.

Evidence worth retaining

Temperature/humidity logs, excursion investigations, layout controls and method-specific limits.

> **Auditor's lens**
>
> Common weakness: Recording room temperature without acting on excursions.
> Question to ask: Would a result still be valid at the highest temperature recorded yesterday?

6.4 Equipment

Equipment includes instruments, software, reference materials, reagents, standards and supporting systems. Fitness for intended use must be established before release and after significant changes.

How to make it work

Maintain equipment identity, status, maintenance and calibration schedules. Define intermediate checks and acceptance criteria. Evaluate prior results when equipment is found out of tolerance.

Evidence worth retaining

Equipment history, certificates, verification results, check charts, out-of-service records and impact reviews.

Auditor’s lens

Common weakness: Filing a calibration certificate without reviewing as-found data or suitability.
Question to ask: Which results could have been affected before the fault was discovered?

6.5 Metrological traceability

Traceability belongs to a measurement result and depends on an evidenced chain of comparisons, with associated uncertainty. An accredited supplier’s logo alone is not a substitute for suitability review.

How to make it work

Map each critical measurand to a traceability route. Check calibration scope, range, uncertainty and dates; assess internal calibration capability and reference materials.

Evidence worth retaining

Traceability map, appropriate certificates, reference-material documentation and uncertainty links.

Auditor’s lens

Common weakness: Using a calibration certificate outside the supplier’s accredited range.
Question to ask: Can we follow this result to a suitable reference without a missing link?

6.6 Externally provided products and services

External laboratories, PT providers, calibration services, reference materials and software may affect results. Purchasing approval must therefore be technical as well as commercial.

How to make it work

Define acceptance criteria, review supplier scope and capability, communicate technical specifications and periodically re-evaluate performance.

Evidence worth retaining

Supplier evaluations, purchasing specifications, incoming acceptance and performance reviews.

Auditor’s lens

Common weakness: Choosing a calibration supplier on price without checking uncertainty capability.
Question to ask: How do we know the purchased service is suitable for this method?

7.1 Review of requests, tenders and contracts

Technical review before acceptance prevents a laboratory promising work it cannot validly perform. It also addresses method, scope, turnaround, reporting and conformity decisions.

How to make it work

Confirm measurand, matrix, range, method, sampling responsibilities, decision rule, uncertainty expectations, subcontracting and reporting format. Record changes and customer agreement.

Evidence worth retaining

Contract review record, deviations, agreed decision rule and communications.

Auditor’s lens

Common weakness: Accepting a pass/fail request without agreeing how uncertainty will be treated.
Question to ask: What exactly did the customer ask us to decide?

7.2.1 Method selection and verification

Using a published standard method does not remove the need to show that the laboratory can perform it properly. Verification tests implementation capability.

How to make it work

Check equipment, operators, matrix, range, detection capability, precision and quality controls against intended use. Control method versions and deviations.

Evidence worth retaining

Verification plan, acceptance criteria, performance data, approval and change record.

Auditor’s lens

Common weakness: Assuming a published method works unchanged on every sample matrix.
Question to ask: What local evidence demonstrates we can achieve the claimed performance?

7.2.2 Method validation

Validation establishes fitness for the intended use, especially for laboratory-developed, non-standard or modified methods. Its extent must match risk and application.

How to make it work

Plan studies for relevant characteristics: bias, precision, selectivity, working range, detection limits, robustness and uncertainty. Set acceptance criteria before testing and review the evidence.

Evidence worth retaining

Validation protocol, raw results, statistical analysis, deviations and approval.

Auditor’s lens

Common weakness: Producing a validation report without predefined success criteria.
Question to ask: Would this method still work for the most difficult sample we accept?

7.3 Sampling

When sampling is part of laboratory activity, the sampling plan and execution can dominate uncertainty and representativeness.

How to make it work

Define location, selection strategy, number, preservation, field conditions, personnel and deviations. Make the chain of custody auditable.

Evidence worth retaining

Sampling plan, field notes, location records, environmental conditions and transfers.

Auditor’s lens

Common weakness: Reporting laboratory precision while ignoring a biased sampling design.
Question to ask: Does the collected material genuinely represent the population of interest?

7.4 Handling of items

A technically excellent measurement cannot rescue a misidentified, damaged or contaminated item.

How to make it work

Use unique identifiers, receipt acceptance, preservation, storage limits, segregation, tracking and disposal arrangements. Record customer decisions on unsuitable items.

Evidence worth retaining

Chain of custody, receipt photos, condition checks, storage logs and deviations.

Auditor’s lens

Common weakness: Relabelling samples without a traceable correction history.
Question to ask: Can we prove the reported result belongs to this exact item?

7.5 Technical records

Technical records must let another competent person reconstruct the work, including who did it, what changed, which equipment was used and how the result was derived.

How to make it work

Capture original observations promptly; preserve audit trails and calculation versions; record corrections without erasing original data.

Evidence worth retaining

Raw instrument output, worksheets, calculation files, signatures, timestamps and audit logs.

> **Auditor’s lens**
>
> Common weakness: Transcribing data into a summary sheet and discarding the original record.
> Question to ask: Could an independent analyst reproduce this result from the retained file?

7.6 Measurement uncertainty

Uncertainty is a reasoned description of dispersion in the value attributed to a measurand. It is not the same as error or instrument tolerance.

How to make it work

Identify meaningful contributors, choose an appropriate model, quantify inputs, combine standard uncertainties and justify coverage. Testing laboratories may need a justified alternative approach where rigorous evaluation is impracticable.

Evidence worth retaining

Uncertainty budget, input sources, sensitivity assumptions, review triggers and reporting rules.

> **Auditor’s lens**
>
> Common weakness: Copying a generic uncertainty budget without matching the actual method.
> Question to ask: Which contributor would most change the reported uncertainty if it doubled?

7.7 Ensuring validity of results

The laboratory needs ongoing independent signals that its process remains under control, rather than waiting for customer complaints or annual audits.

How to make it work

Use control samples, blanks, duplicates, reference materials, PT/ILC, repeat tests, trend charts and peer checks as relevant. Set criteria and responses before a failure occurs.

Evidence worth retaining

QC charts, PT plans/results, investigations, trend review and action effectiveness.

Auditor’s lens

Common weakness: Completing PT participation but not investigating a poor result.
Question to ask: What would tell us tomorrow that today’s results were wrong?

7.8.1–7.8.5 Reporting results

A report must communicate what was measured, by whom, using which relevant method, under what limitations, with clear attribution to the tested or calibrated item. Specific fields differ for tests, calibration and sampling.

How to make it work

Use controlled report templates, verify technical and customer details, distinguish results from opinions, identify outsourced results and check relevant uncertainty statements.

Evidence worth retaining

Approved report, authorisation record, technical review and traceable amendments.

Auditor’s lens

Common weakness: Issuing an accredited-looking report for work outside the accredited scope.
Question to ask: Could a customer misunderstand the scope or limitations of this result?

7.8.6 Statements of conformity and decision rules

A pass/fail statement is a decision, not merely a comparison of a number with a limit. The agreed rule defines how uncertainty affects the risk of a wrong decision.

How to make it work

Identify the specification and decision rule during contract review, unless prescribed. Apply the rule consistently and state it where required. Consider false acceptance and false rejection risks.

Evidence worth retaining

Customer agreement, rule definition, calculation and report wording.

Auditor’s lens

Common weakness: Changing the guard band after seeing a borderline result.
Question to ask: Would another reviewer reach the same conformity decision from the same data?

7.8.7–7.8.8 Opinions, interpretations and report changes

Professional judgement requires authorised personnel and an identifiable technical basis. Amended reports must preserve traceability to the original issue.

How to make it work

Authorise interpretations by competence, distinguish interpretation from raw result, and use controlled revision identifiers and reasons.

Evidence worth retaining

Interpretation approval, amended report history and customer communication.

Auditor’s lens

Common weakness: Replacing a report PDF without recording the original or reason.
Question to ask: Can the customer tell what changed and why?

7.9 Complaints

Complaints can reveal technical weaknesses that normal quality checks missed. Their investigation needs independence and timely communication.

How to make it work

Record receipt, evaluate relevance, assign independent review where feasible, investigate and communicate conclusions. Escalate technical concerns to result impact review.

Evidence worth retaining

Complaint log, investigation, response and trend analysis.

Auditor’s lens

Common weakness: Closing a complaint as “customer misunderstanding” without reviewing evidence.
Question to ask: Did the complaint expose a risk to other reported results?

7.10 Nonconforming work

Nonconforming work includes any laboratory activity or result that fails the laboratory’s own procedures or agreed requirements. The first priority is containment and technical impact.

How to make it work

Stop or restrict affected work, assess significance, decide on report recall or customer notification, authorise resumption and initiate corrective action where indicated.

Evidence worth retaining

Event record, impacted job list, technical risk evaluation, communications and release approval.

Auditor’s lens

Common weakness: Correcting the current worksheet without examining previous affected batches.
Question to ask: How far back could this failure have affected customer results?

7.11 Data and information management

LIMS, spreadsheets, instrument interfaces and cloud platforms are part of the measurement system. Software integrity and access control are technical issues.

How to make it work

Validate relevant calculations and interfaces, control changes, protect audit trails, back up data, test restoration and manage user permissions.

Evidence worth retaining

Validation tests, change tickets, access review, backups and recovery test.

Auditor’s lens

Common weakness: Trusting an unprotected spreadsheet because it was created by an expert.
Question to ask: What happens if a formula is changed silently or a server is unavailable?

8.1 Management system options

Option A provides explicit management-system elements within ISO/IEC 17025. Option B allows use of a management system aligned with ISO 9001, but does not remove technical obligations.

How to make it work

Select and document the route used; ensure all laboratory-specific controls and objective evidence remain demonstrable.

Evidence worth retaining

System map, process ownership, applicable policies and internal audit evidence.

Auditor’s lens

Common weakness: Assuming an ISO 9001 certificate automatically proves ISO/IEC 17025 competence.
Question to ask: Can we demonstrate both management-system effectiveness and technical competence?

8.2–8.4 Documentation and records

Documentation should make work repeatable; records should make performance demonstrable. Their level of detail depends on technical complexity and risk.

How to make it work

Control approval, version, distribution and retention. Separate controlled instructions from generated technical evidence and safeguard electronic records.

Evidence worth retaining

Document register, revision history, retention schedule and access controls.

Auditor’s lens

Common weakness: Keeping superseded instructions available at the bench.
Question to ask: Which instruction was in force when this test was performed?

8.5–8.6 Risks, opportunities and improvement

Risk-based thinking helps the laboratory prioritise threats to valid results and identify better ways of working. It is not limited to maintaining a risk register.

How to make it work

Connect risks to technical controls, assign owners, monitor triggers and evaluate whether changes improve outcomes.

Evidence worth retaining

Risk decisions, improvement projects, indicators and review minutes.

Auditor’s lens

Common weakness: A colourful risk matrix with no link to real controls.
Question to ask: Which risk changed a technical decision this month?

8.7 Corrective action

Effective corrective action addresses why a problem occurred and why existing controls failed to detect or prevent it.

How to make it work

Contain first; investigate root causes proportionately; implement changes; test effectiveness using actual performance evidence.

Evidence worth retaining

Cause analysis, action record, verification data and recurrence monitoring.

Auditor’s lens

Common weakness: Closing actions when training has been delivered, without testing its effectiveness.
Question to ask: What evidence shows this failure is less likely to recur?

8.8 Internal audits

A useful laboratory audit follows a real job through competence, method, equipment, data, uncertainty and report—not just a clause checklist.

How to make it work

Plan risk-based audits; ensure suitable independence and technical competence; sample real work and follow findings to closure.

Evidence worth retaining

Audit programme, working papers, sampled job trail and corrective action follow-up.

Auditor’s lens

Common weakness: An auditor who checks document titles but never examines raw data.
Question to ask: Could we reconstruct one released result from request to report?

8.9 Management review

Management review is a decision forum about laboratory capability, integrity, performance and resourcing. It should result in owned decisions.

How to make it work

Consider performance, audits, complaints, PT, validity trends, workload, competence, risks, changes and resource needs; record decisions and check completion.

Evidence worth retaining

Input pack, decisions, owners, deadlines and follow-up evidence.

Auditor’s lens

Common weakness: A meeting that repeats KPIs but approves no decisions.
Question to ask: What did leadership change because of this review?

Annex A Metrological traceability in practice

Traceability is best understood as a connected measurement chain supported by uncertainty and appropriate reference standards.

How to make it work

Build a one-page chain for one critical measurement. Verify each certificate, scope, range and uncertainty, including in-house links.

Evidence worth retaining

Traceability diagram and certificate cross-reference.

Auditor’s lens

Common weakness: Mistaking a supplier’s general accreditation for coverage of the required measurand.
Question to ask: Where is the weakest link in this measurement chain?

Annex B Management system options in practice

Both routes should support reliable laboratory operations. Neither route excuses gaps in technical competence or record integrity.

How to make it work

Use a process map to show how management controls support technical work.

Evidence worth retaining

Process map and clause-to-evidence matrix.

Auditor’s lens

Common weakness: Treating management-system conformity as a substitute for measurement assurance.
Question to ask: How does the management system detect technically invalid results?

09 Applied examples and audit trails

Example A | Out-of-tolerance micrometer

A micrometer returns from calibration with as-found error outside the laboratory’s acceptance limit. The correct response is not simply to label it and send it for adjustment. Identify the last satisfactory intermediate check, trace affected jobs, evaluate measurement sensitivity, determine whether reported decisions could change and document customer notification where warranted. Resume use only after suitability is confirmed.

Example B | Borderline conformity decision

A measured dimension is 10.02 mm against an upper limit of 10.00 mm, with expanded uncertainty 0.04 mm. The number alone does not determine which conformity statement is appropriate. The laboratory applies the customer-agreed decision rule, records the calculation and communicates any resulting limitation. A rule must not be selected retrospectively to obtain a preferred outcome.

Example C | Method verification

A laboratory introduces a published hardness test method. It checks instrument suitability, reference blocks, operator repeatability, test range, environmental conditions and acceptance criteria. Only after reviewing evidence does the technical authority release the method for routine work.

Example D | PT failure

A proficiency testing result falls outside the scheme criterion. Quarantine assumptions, not merely the score: examine calculations, equipment, analyst technique, sample handling, method bias and other customer work. Record corrective action and demonstrate regained control.

The five-step job-trace audit

1. Select one recently issued report and find the accepted customer request.
2. Trace the named analyst to competence and authorisation records.
3. Verify method revision, equipment status, traceability and environment.
4. Reconstruct raw data, calculation, uncertainty and validity checks.
5. Confirm independent review, decision rule, report approval and any amendments.

10 Visual posters | print-ready field reminders

POSTER 01 | THE CHAIN OF CONFIDENCE

01	REQUEST
02	PEOPLE
03	METHOD
04	EQUIPMENT
05	RAW DATA
06	VALIDITY
07	REPORT

REMEMBER

A result is only as defensible as its weakest link.

POSTER 02 | COMPETENCE IS DEMONSTRATED

01	DEFINE ROLE
02	TRAIN
03	OBSERVE
04	ASSESS
05	AUTHORISE
06	MONITOR
07	REAUTHORISE

REMEMBER

Course attendance is not the same as technical competence.

POSTER 03 | TRACEABILITY

01	MEASURAND
02	REFERENCE
03	CALIBRATION
04	UNCERTAINTY
05	CHAIN
06	REVIEW
07	RESULT

REMEMBER

Check the range and uncertainty, not only the certificate logo.

POSTER 04 | BEFORE YOU SAY PASS

01	SPECIFICATION
02	MEASURE
03	UNCERTAINTY
04	DECISION RULE
05	APPLY
06	REVIEW
07	REPORT

REMEMBER

Agree the decision rule before the result is known.

POSTER 05 | WHEN WORK GOES WRONG

01	STOP
02	CONTAIN
03	TRACE
04	ASSESS IMPACT
05	NOTIFY
06	CORRECT
07	VERIFY

REMEMBER

Contain the technical risk before closing the paperwork.

POSTER 06 | AUDIT ONE REAL JOB

01	CONTRACT
02	AUTHORITY
03	METHOD
04	EQUIPMENT
05	DATA
06	CHECK
07	REPORT

REMEMBER

Follow evidence through the complete laboratory process.

11 Practical templates

Template 1 | Method release record

Method / revision: _____ Scope and matrix: _____
Equipment and reference standards: _____
Verification or validation plan: _____
Acceptance criteria and actual evidence: _____
Uncertainty and QC arrangements: _____
Authorised by / date: _____

Template 2 | Technical result-impact assessment

Trigger / affected instrument or method: _____
Last known acceptable condition: _____
Affected jobs and customers: _____
Worst-case technical impact and uncertainty: _____
Containment and customer communication: _____
Decision / technical approver / resumption criteria: _____

Template 3 | Internal audit job-trace sheet

Job and report number: _____
Contract review and decision rule: _____
Operator competence: _____
Method, equipment and traceability: _____
Raw data, calculations, QC and uncertainty: _____
Report review, release and findings: _____

Final publication-quality self-check

- All clause headings and relevant subclauses are represented in original commentary.
- Examples are labelled illustrative and do not claim to be normative text.
- Accreditation-body-specific rules are separated from ISO/IEC 17025 requirements.
- Readers are directed to the licensed standard and current applicable accreditation rules.
- No official ISO/IEC or accreditation-body endorsement is implied.
- Technical calculations and field examples require discipline-specific validation.

12 Reference framework and notes

Primary normative reference: ISO/IEC 17025:2017, General requirements for the competence of testing and calibration laboratories (consult the licensed edition).

Related technical guidance: JCGM 100 (GUM); JCGM 200 (VIM); relevant ILAC policies on proficiency testing, metrological traceability and uncertainty; ISO/IEC 17043; ISO 17034. Check current editions and applicability.

Supplementary examples reviewed for this edition: ANAB AR 2250 (testing laboratories, effective March 2026); AR 2251 (calibration laboratories, March 2026); AR 2257 (sampling organisations, May 2024); AR 2259 (dimensional measurement laboratories, March 2026). These are ANAB programme-specific documents, not universal ISO/IEC 17025 requirements.

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01 | Impartiality as a controlled operational risk

TECHNICAL APPLICATION · SUPPORTS CLAUSE 4.1

A declaration is an input, not the control itself. The laboratory should be able to demonstrate a sequence of identification, decision, segregation and verification for actual threats to objective results. The standard does not prescribe a risk matrix or require a particular declaration form; the operating model below is an RPG Excellence implementation method.

Decision gate	Responsible role	Record / pass criterion
Who can influence the outcome?	Quality lead + technical manager	Influence map covers quotation, item selection, method, calculation, interpretation, conformity, review and release.
Is any connection material?	Employee / contractor + reviewer	Event-based disclosure states the relationship and affected jobs; avoid intrusive information unrelated to work.
Could bias enter this job?	Quality lead + impartial decision-maker	Identify the exact route: sample allocation, re-testing, editing, suppression, conformity judgement or publication.
Can the risk be controlled?	Unaffected manager	Reassignment or independent technical decision has actual authority, access control and documented approval.
Did the control operate?	Independent verifier	LIMS logs, assignments, review evidence and released report agree with approved safeguard.

Worked case: commercial relationship with a customer

Situation. A technician is asked to test weld specimens for a customer whose project manager is their spouse. A result below the acceptance threshold could affect the customer's contract.

Practical response. The technician reports the relationship before allocation. The quality lead assesses the affected tasks, assigns another competent technician and independent reviewer, removes the original technician from report approval, and verifies LIMS permissions. If separation cannot be demonstrated, the laboratory declines or reallocates the work.

Objective evidence. Declaration; job-risk decision; task allocation; permission history; raw-data reviewer; signatory authorisation; verification after release.

> **Assessor challenge:** What evidence shows this specific job was independent, rather than merely covered by a signed annual policy?

Worked case: production laboratory reporting to operations

Situation. A laboratory embedded in a manufacturing business is pressured by a production manager to repeat marginal tests until a satisfactory outcome appears.

Practical response. Separate commercial and production objectives from method selection and technical disposition. Define how valid repeat testing is justified; require investigation and retention of every result; grant a technical authority the right to block inappropriate report release. Escalate threats to senior management outside the interested chain.

Objective evidence. Documented reporting lines; quality escalation policy; test sequence logs; all repeat-test justifications; nonconforming-work records; management responses.

> **Assessor challenge:** Can you show an occasion where a result was retained or reported despite pressure to change it?

Illustrative scoring method — a management tool, not an ISO scoring requirement

Likelihood	Impact on confidence	Decision
Low: influence route remote; controls proven	Limited, recoverable effect	Document safeguard, verify at review.
Plausible route to the technical decision	Could alter conformity or reported result	Reassign the decision or provide demonstrably independent control.
Direct financial/personal interest and uncontrolled ability to amend or approve	Result credibility materially compromised	Stop allocation/release until exposure is removed; decline work when necessary.

Do not combine scores into an apparently precise number unless the laboratory can explain the meaning of the scales. Residual risk is a decision supported by evidence; it is not automatically “low” because a control is written down.

02 | Confidentiality, access and digital disclosure

TECHNICAL APPLICATION · SUPPORTS CLAUSES 4.2, 7.11

Design confidentiality as a route-specific control: identify what enters the laboratory, who is authorised to see it, what is shared with suppliers, and how release is authenticated. Legal release may be necessary; assess notification requirements before disclosure. Maintain access according to role and review permissions after personnel changes.

Information / event	Preventive control	Detection and recovery
Customer design drawings	Classify project; restricted project space; supplier nondisclosure terms	Quarterly account review; export log; revoke access at job close.
Instrument raw-data and audit trails	Named accounts, changes attributable to persons, backup and restore tests	Audit event history and integrity checks after anomalous results.
Reports transmitted outside laboratory	Verified recipient; approved channel; retained issue version	Distribution history; incident procedure for misdirected report.
Cloud-hosted LIMS	Vendor due diligence and contractual data responsibilities	Access testing, availability/restore drills, change log and incident notification.

Worked case: corrected report after issue

Situation. A report is emailed to a customer and later a revised result is approved. Both versions remain accessible through different shared-folder locations.

Practical response. Issue the amendment under a unique reference linking it to the original; make the superseded report clearly identifiable, control future access and notify the customer through an agreed route. Investigate whether other records and conformity statements rely on the original.

Objective evidence. Report revision trail; amendment reason; authorised review; customer notification; access log.

> **Assessor challenge:** Which report is currently authoritative, and can a recipient distinguish it from the superseded version?

03 | Competence, authorisation and witness assessment

TECHNICAL APPLICATION · SUPPORTS CLAUSES 5, 6.2

Training attendance is not an authorisation. For every task that can influence result validity, define the technical judgement required, assess performance against known criteria and record the exact scope of permission. A technician may be competent to run a test but not to validate the method, interpret a nonstandard outcome or sign a report.

Stage	How to carry it out	Acceptance evidence
Define	List methods and critical activities by equipment, range, site and level of complexity.	Competence standard names required technical knowledge, practical skill and deviation judgement.
Train / familiarise	Observe methods, calculations, records, decision rules and escalation.	Training log + version of SOP + practice materials.
Assess	Witness a live/representative test; challenge abnormal data and calculate an independent check.	Observed assessment and correctly identified stop conditions.
Authorise	Technical authority approves method, range, tasks and release permissions.	Dated permission in an accessible role-based authorisation matrix.
Monitor	Review QC, PT, errors, infrequent tasks, revisions and absence.	Reassessment trigger, periodic evidence sample and restriction where necessary.

Worked case: tensile testing at two sites

Situation. An experienced technician transfers to a site using a different extensometer and specimen preparation process. Their previous authorisation is for a different machine and method configuration.

Practical response. Provide site-specific familiarisation; witness calibration/verification checks, specimen alignment, strain calculation and treatment of fracture outside the gauge length. Compare accepted reference data and authorise only the applicable method and equipment.

Objective evidence. Site-specific witness record, example test report, raw stress-strain trace, competence decision and authorised scope.

> **Assessor challenge:** Where is the evidence that the individual can recognise and correctly handle a technically invalid fracture?

Sample assessment questions

- What raw observation would require stopping the method before a result is calculated?
- How would you detect a transposed unit or a spreadsheet formula error?
- What changes if the instrument is operating near the accredited range limit?
- Which part of the reported result is outside your personal authority?

04 | Environment, equipment and in-house calibration

TECHNICAL APPLICATION · SUPPORTS CLAUSES 6.3–6.4

Define environmental criticality from the method rather than using one arbitrary limit for every laboratory. Identify the measurand, dominant environmental influences, applicable limits, monitoring frequency and the decision when conditions fall outside control. Equipment management must link fitness for use to actual results, not just a due-date sticker.

Lifecycle stage	Operational control	Disqualifying example
Selection and acceptance	User requirement; range, accuracy, resolution, uncertainty and suitability.	A balance resolves 0.1 g where method uncertainty requires 0.001 g.
Commissioning	Identification; installation; calibration/verification; software setup; method checks.	Equipment placed in service on delivery, before checks.
Routine use	Intermediate checks; environmental records; authorised operator; preventive maintenance.	Out-of-control check treated as a maintenance note only.
Out-of-tolerance event	Quarantine equipment; identify last known acceptable status; assess affected result population.	Calibration error discovered but no evaluation of previously issued results.
Return to service	Repair, calibration, verification and technical authorisation.	Sticker replaced without confirming intended-use performance.

Worked case: out-of-tolerance reference thermometer

Situation. An external calibration shows a reference thermometer reading 0.8 °C high at a laboratory control point. It has supported an ageing test chamber for eight months.

Practical response. Determine the correction history and uncertainty, identify all affected work since the last satisfactory check, assess whether measured chamber conditions breached the method tolerance and whether reported conclusions change. Retest retained specimens if technically justifiable; amend or withdraw results when necessary.

Objective evidence. Calibration history; use log; impact assessment by job; customer decisions; restart authorisation.

> **Assessor challenge:** How was the starting point of the affected period chosen, and why is it defensible?

In-house calibration: minimum defensible evidence

• Document the calibration method, reference standards, range, measurand, environmental conditions and measurement model.

• Demonstrate staff competence for the calibration itself, not solely for operation of the item.

• Establish traceability of reference standards and evaluate uncertainty, including correlations where material.

• Retain as-found and as-left results, acceptance criteria, any adjustment, conclusion and independent verification appropriate to risk.

• Verify the chosen interval using drift, checks, use and consequence of failure; interval optimisation is not a substitute for valid uncertainty evaluation.

05 | Metrological traceability and technical purchasing

TECHNICAL APPLICATION · SUPPORTS CLAUSES 6.5–6.6; ANNEX A

Traceability belongs to the measurement result, not to a sticker, instrument or ISO certificate alone. A laboratory needs a documented chain with suitable uncertainties and technical applicability. Review the specific calibration scope, measurement points, uncertainties and status of the provider rather than assuming every service from an accredited organisation is accredited.

Check	Question	Evidence
Measurand and range	Is the calibration appropriate to the measurement being made?	Certificate points, range, conditions, correction use.
Source	Is the provider's accredited capability appropriate?	Applicable scope and status; NMI or other suitable traceability source.
Uncertainty	Are the reference and laboratory uncertainties fit for intended use?	Budget, reference certificate, decision on fitness.
Transcription	How does the laboratory apply corrections and safeguard data?	Controlled spreadsheet/LIMS constants, independent verification.
Change	What if the reference has drifted or accreditation status changes?	Change control and impact review.

Worked case: calibration supplier has accreditation but not for required range

Situation. A supplier holds ISO/IEC 17025 accreditation for mass but the certificate used by the laboratory concerns a pressure gauge at a range outside the provider's published accredited scope.

Practical response. Stop using the accreditation claim as evidence for that pressure calibration. Review technical competence and traceability against relevant policy, arrange a suitable accredited calibration if needed, and evaluate any already-issued results that relied on the certificate.

Objective evidence. Supplier scope check, revised qualification, calibration certificate, affected-work review and decision.

> **Assessor challenge:** Which line in the supplier's scope matches this measurand, range and capability?

06 | Contract review and accredited scope control

TECHNICAL APPLICATION · SUPPORTS CLAUSES 5.3, 7.1

Contract review should occur before the laboratory accepts the work. Translate the customer question into measurand, sample matrix, method, specification, required uncertainty, sampling responsibility, turnaround, decision rule, subcontracting permission and report claim. Where a customer's requested method cannot meet the purpose, communicate and record the discrepancy before proceeding.

Customer request	Review decision	Potential consequence
"Please certify product passes."	Define rule, specification edition, uncertainty treatment, and whether the laboratory may issue a conformity statement.	A post-test decision rule biases the decision.
"Test in the next 24 hours."	Verify sample holding time, equipment availability and trained authorisation.	Speed compromises valid sample handling or controls.
"Use accredited method."	Check actual methods, matrices, locations and ranges on schedule.	Inaccurate claim of accreditation.
"Subcontract the overflow."	Review customer consent and externally provided work controls as applicable.	Undisclosed or misrepresented laboratory activity.

Worked case: unsupported scope claim

Situation. The laboratory advertises "all corrosion testing accredited", while the published scope covers only one method in a specified matrix.

Practical response. Correct marketing and quotations immediately; review open contracts and past reports for misleading claims; describe accredited and non-accredited activities separately and create a release checkpoint against the scope schedule.

Objective evidence. Scope-to-service mapping, amended collateral, contract reviews, customer communication where relevant.

> **Assessor challenge:** Can commercial staff demonstrate how they determine whether a particular order is within the accredited scope?

07 | Method verification and validation: fitness for purpose

TECHNICAL APPLICATION · SUPPORTS CLAUSE 7.2

Verification establishes that the laboratory can perform an existing method to required performance. Validation determines whether a method, including a modified or laboratory-developed method, is suitable for its intended use. Neither is satisfied by a method title and a training sign-off. Plan acceptance criteria before collecting data. The 2025 Eurachem method-validation guide provides modern experimental-design guidance; apply it within the discipline, specification and customer context.

Characteristic	Illustrative study	Decision before start
Repeatability	Repeat independently prepared specimens across meaningful concentration or load range.	Set maximum acceptable repeatability SD or CV from intended decision.
Intermediate precision	Vary operator, day, equipment or batch as technically relevant.	Define allowable between-condition variation.
Bias / trueness	Reference material or recovery against defensible assigned value.	Define allowable bias, including its uncertainty.
Selectivity	Compare plausible interferents or matrix effects.	Specify allowable influence at the decision level.
Working range	Check response and performance across intended range.	Define fit-to-model and maximum residual/error.
LOD / LOQ	Use matrix blanks or low-level samples where relevant.	Select statistical approach and intended reporting purpose.
Robustness	Challenge realistic variation (temperature, time, preparation).	Define tolerable sensitivity and method controls.

Worked case: standard chemical method transferred to a new matrix

Situation. A laboratory uses a published method validated for clean water but now receives industrial effluent containing interfering ions.

Practical response. Define matrix and customer requirement; run interference/recovery, precision and quantification studies at decision concentrations. Treat use beyond documented applicability as requiring additional validation rather than relying on the original publication alone. Approve the matrix extension before routine release.

Objective evidence. Validation plan; data and statistics; acceptance decisions; authorised method scope; QC plan update.

> **Assessor challenge:** What evidence establishes the method’s validity specifically for this matrix and concentration range?

Illustrative precision calculation

Six replicate results (units): 10.01, 10.04, 9.98, 10.02, 9.99, 10.03. Mean = 10.0117; sample standard deviation = approximately 0.0232; relative standard deviation = approximately 0.23%. These statistics alone do not prove the method is fit for purpose: the acceptance target and relevant bias, range and sampling influences must also be considered.

08 | Sampling and handling: protect representativeness

TECHNICAL APPLICATION · SUPPORTS CLAUSES 7.3–7.4

Sampling quality may dominate the uncertainty of a result, particularly when the material is heterogeneous. A laboratory must be clear about whether sampling is within its activities and accreditation scope. Document the selection method, number and position of increments, equipment, conditions, preservation and deviations; protect identification and chain-of-custody through receipt, storage, analysis and disposal.

Worked case: batch sampling hides variability

Situation. A client supplies a mixed-material batch. A sampler takes one visually convenient scoop even though the plan specifies spatial increments. The reported concentration is near a legal limit.

Practical response. Treat the deviation as a potential representativeness failure. Investigate whether the original sample can support the intended claim; obtain a technically justified representative sample where feasible, document circumstances and notify the customer before issuing conclusions.

Objective evidence. Sampling plan, actual sampling record, deviation, reassessment of uncertainty and report limitation.

> **Assessor challenge:** Does the reported conclusion apply to the batch or only to the material actually collected?

Control point	Useful record
At collection	Sampler; identity; exact point/time; container; conditions; increment and composite pattern; deviations.
At receipt	Condition and integrity check; preservation; transfer record; acceptance/rejection.
During storage	Temperature monitoring, holding time and access control.
At disposal	Release/retention authority; final identification and disposal log.

09 | Measurement uncertainty: show the measurement model

TECHNICAL APPLICATION · SUPPORTS CLAUSES 7.6, 7.8

A credible uncertainty estimate begins with the result equation and the decision being supported. List contributors, classify distributions and data sources, convert to standard uncertainties, assess dependence, combine appropriate variances and select a defensible coverage factor. Be especially alert to double-counting repeatability or calibration contributions already included in validation data.

Worked measurement example: dimensional measurement

Suppose the reported length is 50.020 mm. Independent standard uncertainty components are calibration 0.010 mm, repeatability 0.015 mm, and temperature influence 0.008 mm. Combined standard uncertainty = sqrt($0.010^2 + 0.015^2 + 0.008^2$) = 0.0197 mm (rounded). With k = 2, expanded uncertainty U = 0.0394 mm, reported as approximately 0.04 mm if this rounding is appropriate to the laboratory's reporting policy. The result is illustrative, assumes independence and a suitable approximately 95% coverage interpretation, and does not model every real measurement.

Contribution	Standard uncertainty	Source / verification
Calibration	0.010 mm	Applicable certificate, coverage-factor conversion and correction treatment.
Repeatability	0.015 mm	Representative precision data for the result calculation, not arbitrarily selected.
Temperature	0.008 mm	Measurement model and relevant distribution.
Combined	0.0197 mm	Root-sum-square only if contributions are reasonably uncorrelated.
Expanded, k = 2	0.0394 mm	Coverage choice and distribution assumptions documented.

Practical uncertainty-budget review

• Verify the measurand, units and mathematical measurement model.

• Check that standard uncertainty from a rectangular distribution is calculated as half-width divided by sqrt(3), when that model applies.

• Identify covariance where the same reference or correction affects more than one input.

• Check whether calibration, sampling and operator variation are already represented in the empirical precision estimate.

• Confirm that a reported conformity statement uses the agreed decision rule, not the uncertainty estimate alone.

10 | Quality control, proficiency testing and the validity of results

TECHNICAL APPLICATION · SUPPORTS CLAUSE 7.7

A PT result is one point-in-time external check. It does not replace routine QC, suitable method verification or investigation of drift. Define a risk-based validity plan by method family, range, matrix, complexity, recent performance, throughput and consequence of an erroneous result. ILAC P9 concerns accreditation bodies' PT policy; specific schemes, such as the supplied ANAB documents, may impose numerical frequencies that are not universal ISO/IEC 17025 rules.

Failure risk	Control	Trigger and action
Drift over time	Control chart or intermediate check	Predefined warning/action limits; inspect cause and affected work.
Matrix interference	Matrix spike, CRM or independent method	Bias beyond predefined limit; stop relevant samples and investigate.
Calculation/transcription	Independent calculation, controlled software	Unexplained difference; hold approval.
Between-laboratory discrepancy	PT/ILC selected for scope, range and uncertainty	Unsatisfactory result; investigate both technical and systemic mechanisms.
Infrequent method	Reference challenge / comparison before use	Competence and instrument readiness evidence absent.

Worked case: unsatisfactory PT but routine QC passes

Situation. A laboratory receives a PT z-score of 3.2 for a trace-element method. Routine control samples have remained within the laboratory's established limits.

Practical response. Do not dismiss PT because internal QC passed. Check sample receipt, measurand definition, assigned value, method bias, calibration traceability, matrix effects, transcription, uncertainty and comparability. Decide whether related results require review; update QC where the exercise exposes a blind spot.

Objective evidence. PT report, raw files, investigation, affected-work decision, corrected controls and subsequent verified performance.

> **Assessor challenge:** How did the investigation distinguish a one-off mistake from systematic bias that routine QC could not detect?

11 | Decision rules: results, uncertainty and risk

TECHNICAL APPLICATION · SUPPORTS CLAUSES 7.1, 7.8.6

Before evaluating conformity, identify the specification, the intended statement and the decision rule. Distinguish an acceptance interval from the specification interval. The applicable rule may be specified by a regulator, customer or standard; where not inherent it must be agreed with the customer where applicable. ILAC G8 and Eurachem guidance explain how different rules redistribute risks of false acceptance and false rejection.

Worked upper-limit example

Specification limit: value must not exceed 10.00 units. Result y = 9.98 units and expanded uncertainty U = 0.06 units. Under simple acceptance based only on the measured value (if authorised), y is below the limit. Under a guard-banded rule requiring $y + U \leq 10.00$, the upper bound is 10.04, so this result does not meet that conservative acceptance criterion. That is not the same as proving the actual item is nonconforming. A conditional or inconclusive outcome may be appropriate if the agreed scheme permits it.

Rule example	Calculation	Interpretation
Simple acceptance	$9.98 \leq 10.00$	Accept if the contract/authority specifies that rule; shared risk near the limit.
Guard band using U	$9.98 + 0.06 = 10.04 > 10.00$	Cannot declare pass under this illustrative guard-band rule.
Four-outcome example	Compare the result uncertainty interval with the limit	May allow an “inconclusive” category depending on defined scheme.

Never choose the rule after obtaining the result. Preserve the agreed rule, its revision, the measured result and uncertainty, and the person authorised to issue the statement.

12 | Reporting, technical judgement and amended results

TECHNICAL APPLICATION · SUPPORTS CLAUSES 7.8.1–7.8.8

Treat the report as the final controlled expression of the work. The reader must understand what was tested, by which method, where relevant, and what the result actually means. Keep results, uncertainty, conformity statements, opinions and interpretations visibly distinguishable. Ensure authorised personnel understand the boundary between a measured fact and a technical opinion.

Report decision	Control
Accreditation status	Describe only the accredited activities and permitted symbol/claim.
Uncertainty	Include it when required by the standard, customer, method or relevant decision.
Sampling	Identify sampling responsibilities and any deviations/limitations.
Opinions / interpretations	Identify authorised basis and label appropriately.
Amendment	Link corrected issue to original, identify changed information and maintain release history.
Electronic delivery	Validate recipients, prevent post-approval edits and retain issued copy.

Worked case: opinion presented as test result

Situation. An analyst writes “material is suitable for offshore deployment” beneath tensile and impact results although no service-design assessment or authorisation exists.

Practical response. Separate directly measured values from any interpretation; determine whether a competent technical signatory can make the conclusion using agreed criteria and sufficient scope; otherwise remove the unsupported claim and communicate the limitations.

Objective evidence. Technical review, method scope, opinion authorisation, amended report, customer communication.

> **Assessor challenge:** What technical framework and competence demonstrate that the service suitability opinion was justified?

13 | Nonconforming work and corrective action

TECHNICAL APPLICATION · SUPPORTS CLAUSES 7.10, 8.7

Not every incident requires the same stoppage, but every significant deviation needs an authorised technical disposition proportionate to the risk. Separate immediate control of affected work from root-cause corrective action. A closed action is not an effective action unless evidence demonstrates that the failure mode has been controlled.

Step	Minimum control	Record to retain
Contain	Stop affected process, quarantine item/result, protect records and identify affected work.	Incident, first-failure time, equipment and jobs.
Evaluate significance	Determine likely effect on validity and reported decisions; use worst credible impact where data are weak.	Technical rationale, affected results and risk disposition.
Decide customer action	Inform customer/withdraw/amend/retest when appropriate; authorise notification.	Communication log and decision.
Find causes	Assess technical, training, equipment, method and process contributors.	Evidence-based cause evaluation.
Verify effectiveness	Test revised control on new work; check recurrence and previously affected population.	Follow-up sampling and approval.

Worked case: spreadsheet cell overwritten

Situation. During review, an analyst notices that a protected calculation cell contains a fixed number on a copied worksheet used for 15 reports.

Practical response. Secure the spreadsheet and issue control. Recalculate all affected cases using a verified formula and source raw data; determine which reported conclusions change and notify customers as appropriate. Identify how protection, version control and independent review failed; validate the corrected workbook.

Objective evidence. Original and corrected workbook hashes/versions, impacted report list, independent recalculation, technical release and effectiveness test.

> **Assessor challenge:** How do you know the defect is absent from every copy of the calculation template?

14 | LIMS assurance, data integrity and cybersecurity

TECHNICAL APPLICATION · SUPPORTS CLAUSE 7.11

A LIMS can automate a defective process faster than a paper system. Establish a validation strategy based on intended use, result-critical functions and failure consequences. Define user roles, separation of entry and approval, controlled changes, audit trails, backup/restore and management of external platforms. An audit trail that nobody reviews is a weak detective control.

Risk scenario	Validation test	Routine evidence
Transposed units	Enter reference result and verify automatic unit conversion and report value.	Signed test script; regression test after updates.
Unauthorised reapproval	Attempt to approve with insufficient role and verify denial.	Role matrix; access review.
Post-release modification	Attempt edit after report release; verify version/history preservation.	Audit-trail review; report linkage.
Restore failure	Restore a representative backup in a controlled environment and reconcile raw/derived data.	Restore results and discrepancy log.

Worked case: vendor updates instrument software

Situation. An automatic update changes how a chromatography integration baseline is selected. The software version is new and old processed files are not automatically recalculated.

Practical response. Apply risk-based change control, compare defined test datasets before and after update, confirm integration parameters and calculated results, assess effects on open and previously released work if the change is retrospective, and authorise deployment only after acceptance criteria are met.

Objective evidence. Vendor change record, regression dataset, comparison results, deviation review and release approval.

> **Assessor challenge:** Which result-critical features were deliberately challenged during the upgrade test?

15 | Management system, audits and management review

TECHNICAL APPLICATION · SUPPORTS CLAUSES 8.1–8.9; ANNEX B

Option A and Option B concern management-system arrangements; they do not create alternative technical standards. Under Option B, an ISO 9001-based management system must genuinely support consistent fulfilment of ISO/IEC 17025. Audit planning should cover real tests, methods and risk, not only documentation. Senior review must make resource and policy decisions based on evidence of technical performance.

Vertical audit — one report backwards to its source

Trace step	Sample evidence / challenge
Report	Correct customer, method, result, uncertainty, decision rule and approved signatory.
Review	Technical-review note, raw-data comparison, comments and final authorisation.
Calculations	Controlled tool version, source results, unit and transcription checks.
Equipment	State at time of test, calibration evidence, intermediate checks, environment.
Personnel	Authorisation scope at testing and review dates.
Method	Applicable revision, verification/validation and deviation disposition.
Contract	Request, accepted scope, customer decision rule and subcontracting arrangements.
Governance	Applicable impartiality and confidentiality safeguards.

Management review dashboard: measurements that change decisions

Indicator	Signal	Management response
PT/ILC performance by method	Recurring bias or marginal results	Approve investigation, alternative scheme or method investment.
Overdue equipment and intermediate checks	Growing proportion beyond planned controls	Reallocate maintenance capacity or suspend affected work.
Personnel authorisation gap	Tasks depend on one signatory	Approve training and succession; restrict work as necessary.
Report amendment rate and severity	Rise in result-changing corrections	Targeted system audit and validation of calculations.
Corrective action effectiveness	Repeat failure despite closure	Reopen systemic cause and leadership accountability.

16 | Sector-specific considerations and accreditation-body overlays

TECHNICAL APPLICATION · SUPPORTS ANNEXES A AND B; SUPPLEMENTARY POLICIES

The ANAB documents supplied for testing (AR 2250), calibration (AR 2251), sampling organisations (AR 2257) and dimensional measurement (AR 2259) are supplementary programme requirements. They cannot be treated as universally binding on UKAS, A2LA, DAkkS or other accreditation-body clients. Verify the relevant scheme, version and published scope when using any numeric PT frequency or retention requirement.

Discipline	Technical emphasis	Supplementary check
Testing	Matrix effects, method validation, PT and in-house calibration support.	ANAB AR 2250 where ANAB testing scheme applies.
Calibration	CMC, uncertainty reporting, traceability and calibrations supporting results.	ANAB AR 2251 and ILAC P14 where applicable.
Sampling organisation	Representativeness, conditions, chain of custody and sampling competence.	ANAB AR 2257 for relevant ANAB sampling accreditation.
Dimensional measurements	Measurement strategy, reference dimensions, tolerance/uncertainty decisions.	ANAB AR 2259 for relevant ANAB scope.

The 2009 UNIDO guide is a useful example of practical training, authority mapping, record traceability and audit planning, but its clause numbering and certain mandatory-procedure statements reflect the superseded 2005 edition. Translate useful methods into the requirements and flexibility of ISO/IEC 17025:2017 before recommending them.

17 | Reusable laboratory records and templates

TECHNICAL APPLICATION · SUPPORTS PRACTITIONER TOOLKIT

Template A — Impartiality decision record

Field	Entry required
Risk and affected jobs	Customer/employee relationship or pressure route; which method, reports and activities are exposed.
Decision-maker	An individual who is not personally affected by the conflict.
Controls	Assignment restrictions, independent reviewer, access controls, no discretionary re-test.
Residual decision	Reasons and evidence; conditions of acceptance / refusal.
Effectiveness	Audit log review, job sample, recurrence and event trigger.

Template B — Method verification / validation approval

Field	Entry required
Intended use	Sample matrix, measurand, working range, decision level and customer needs.
Performance objectives	Precision, bias, LOD/LOQ, robustness, uncertainty and other relevant criteria.
Study design	Replicates, operators, concentration levels, days, reference materials, statistics.
Raw data and evaluation	Traceable files, calculations, deviations, observed performance and comparison to targets.
Disposition	Authorised scope, limitations, QC frequency, periodic review and owner.

Template C — Equipment out-of-tolerance impact review

Field	Entry required
Discovery and limits	Reference and actual error; uncertainty; affected measurement conditions.
Affected window	Last known acceptable check through confirmed containment, with rationale.
Job population	All results supported by the instrument, including subcontracted reports where relevant.
Quantified impact	Recalculation, worst credible impact, retest and conclusion change.
External action	Customer notification and amended/withdrawn reports if required.
Release	Approval after repair/calibration/verification and preventive actions.

Template D — Decision rule authorisation

Field	Entry required
Specification	Edition, limit(s), measurand, units, customer/regulatory requirement.
Rule	Simple, guarded, conditional or explicitly prescribed; treatment of uncertainty and risk.
Agreement	Contract record; confirmation prior to work unless inherent in the specification.
Implementation	Controlled calculation, test cases near the limit and authorised report wording.
Review	Owner, revision, change trigger, review of issued outcomes.

Template E — Vertical audit sampling sheet

Question	Evidence and outcome
Request and contract reviewed?	Document ID / sample ID / date / person / finding / conclusion
Impartiality screened?	Document ID / sample ID / date / person / finding / conclusion
Sample traceable?	Document ID / sample ID / date / person / finding / conclusion
Method verified and applicable?	Document ID / sample ID / date / person / finding / conclusion
Equipment and staff authorised?	Document ID / sample ID / date / person / finding / conclusion
QC controls satisfied?	Document ID / sample ID / date / person / finding / conclusion
Calculations and uncertainty checked?	Document ID / sample ID / date / person / finding / conclusion
Report and conformity decision authorised?	Document ID / sample ID / date / person / finding / conclusion

18 | Team posters — critical controls at a glance

TECHNICAL APPLICATION · SUPPORTS PRINTABLE REFERENCE PAGES

POSTER 1 • IMPARTIALITY IS A LIVE RISK

01 DECLARE links and pressures as they arise

02 ASSESS who could influence the result

03 SEPARATE conflicted technical decisions

04 VERIFY actual job-level safeguards

05 REVIEW after role, client or incentive changes

Manager discussion: Which recent laboratory record proves these steps actually happened?

POSTER 2 • NO AUTHORISATION, NO TASK

01 DEFINE competence by method and range

02 WITNESS the work and abnormal decisions

03 RECORD results of practical assessment

04 AUTHORISE exact tasks and signatures

05 REASSESS when methods or performance change

Manager discussion: Which recent laboratory record proves these steps actually happened?

POSTER 3 • FOLLOW THE RESULT

01 REVIEW the customer requirement

02 CHECK method and approved personnel

03 CONFIRM equipment, traceability and conditions

04 REVIEW raw data and independent QC

05 APPLY the agreed rule and release a clear report

Manager discussion: Which recent laboratory record proves these steps actually happened?

POSTER 4 • STOP, CONTAIN, EVALUATE

01 IDENTIFY the deviation

02 PROTECT affected samples and results

03 ASSESS the whole population of work

04 DECIDE re-test, amendment or notification

05 VERIFY corrective action and release authority

Manager discussion: Which recent laboratory record proves these steps actually happened?

19 | Research references and editorial boundaries

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Editorial distinction: ISO/IEC 17025 clauses establish the normative basis. The process maps, numeric examples, templates, audit challenges and posters are original RPG Excellence implementation proposals. ILAC documents, accreditation-body policies and technical guidance must be checked for applicability and changes prior to use. Published standards are copyright-protected and not reproduced here.

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